

PHY 211

Lecture 12 Notes

By

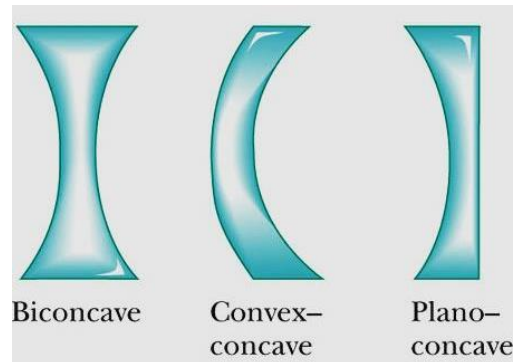
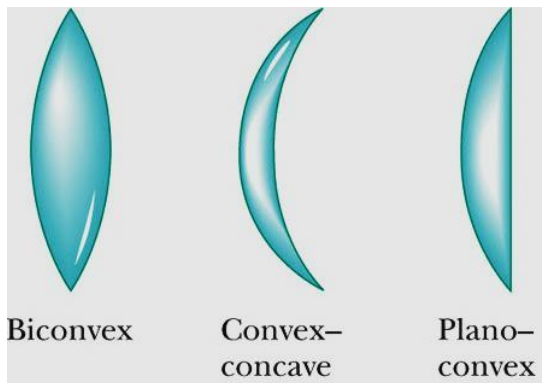
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Fall 2024

Lenses

➤ Thin lenses:

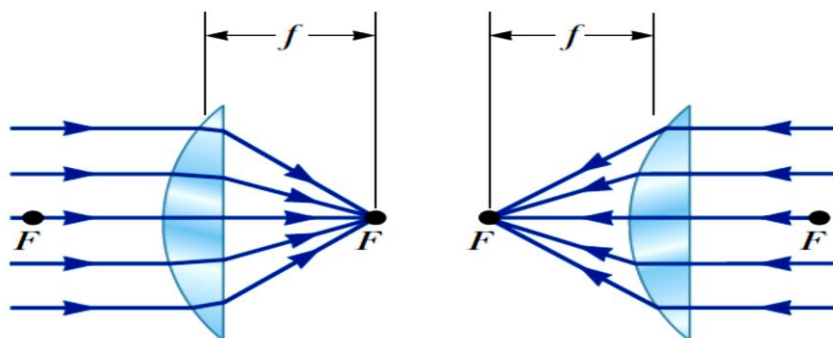
- Types of lenses
1- Spherical lenses 2- Cylindrical lenses 3- Spherical-cylindrical lenses
- **Spherical lens:** It is a part of transparent refracting medium bounded by two spherical surfaces or by one spherical and one plane surface.



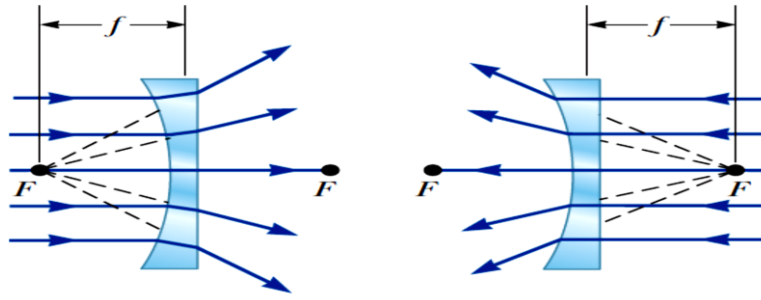
Converging lenses

Diverging lenses

- **Converging lenses:** They have +ve focal lengths and are thicker at the center than at the edges.
- **Diverging lenses:** They have -ve focal lengths and they are thicker at the edges than at the center
- **Focal point of converging length:**



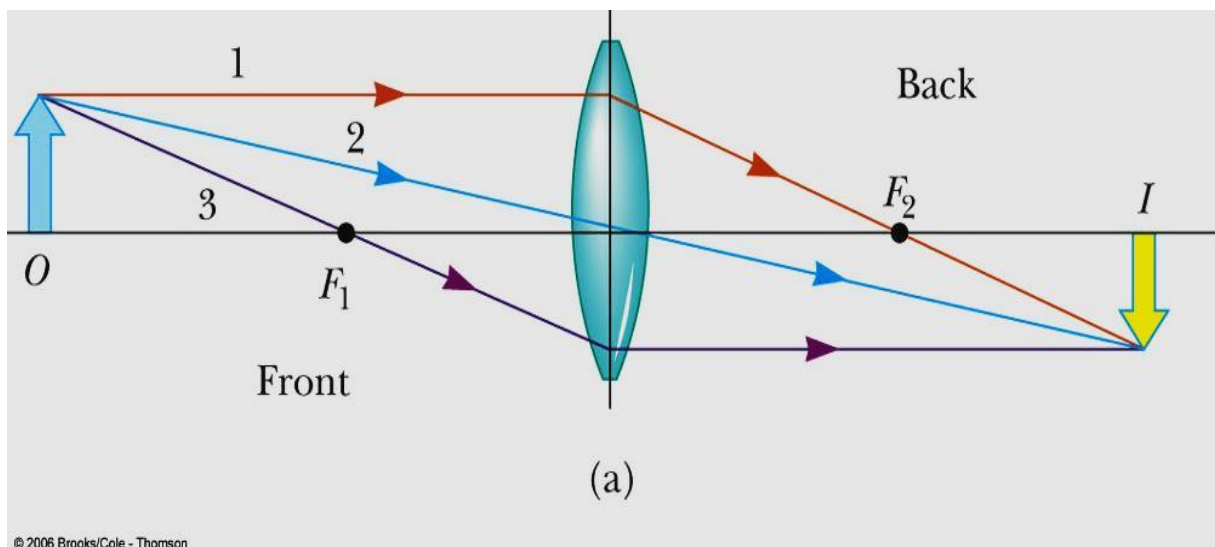
- **Focal Point of diverging length:**



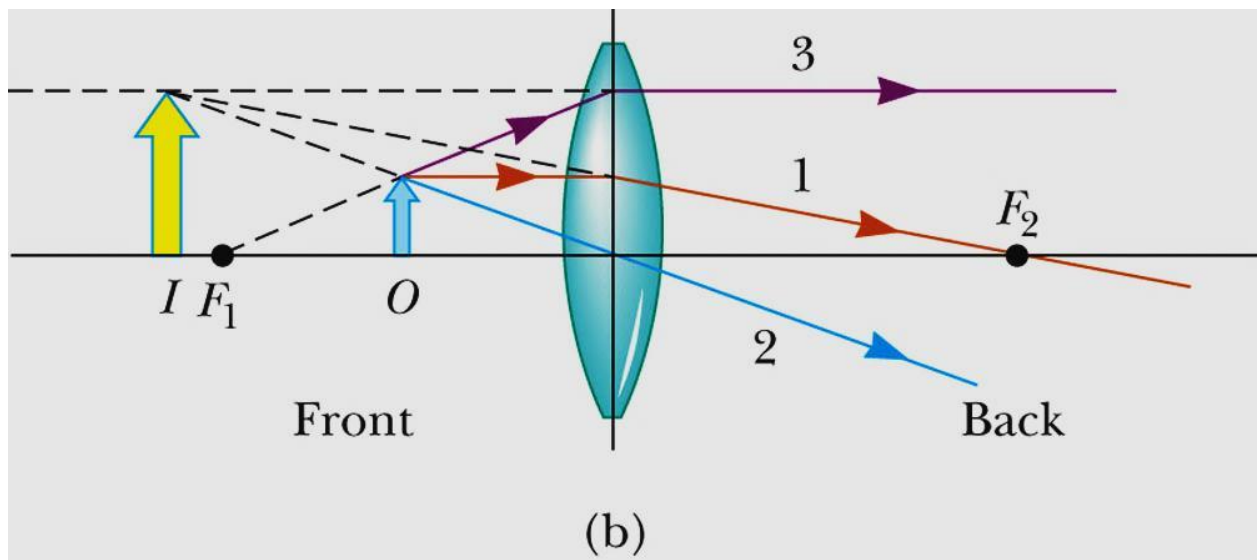
- **Ray diagram of thin lenses:**

- 1- The first ray is drawn parallel to the principal axis. After being refracted by the lens, this ray passes through (or appears to come from) one of the focal points.
- 2- The second ray is drawn through the center of the lens. This ray continues in a straight line.
- 3- The third ray is drawn through the other focal point and emerges from the lens parallel to the principal axis.

- **For Convex lens:**

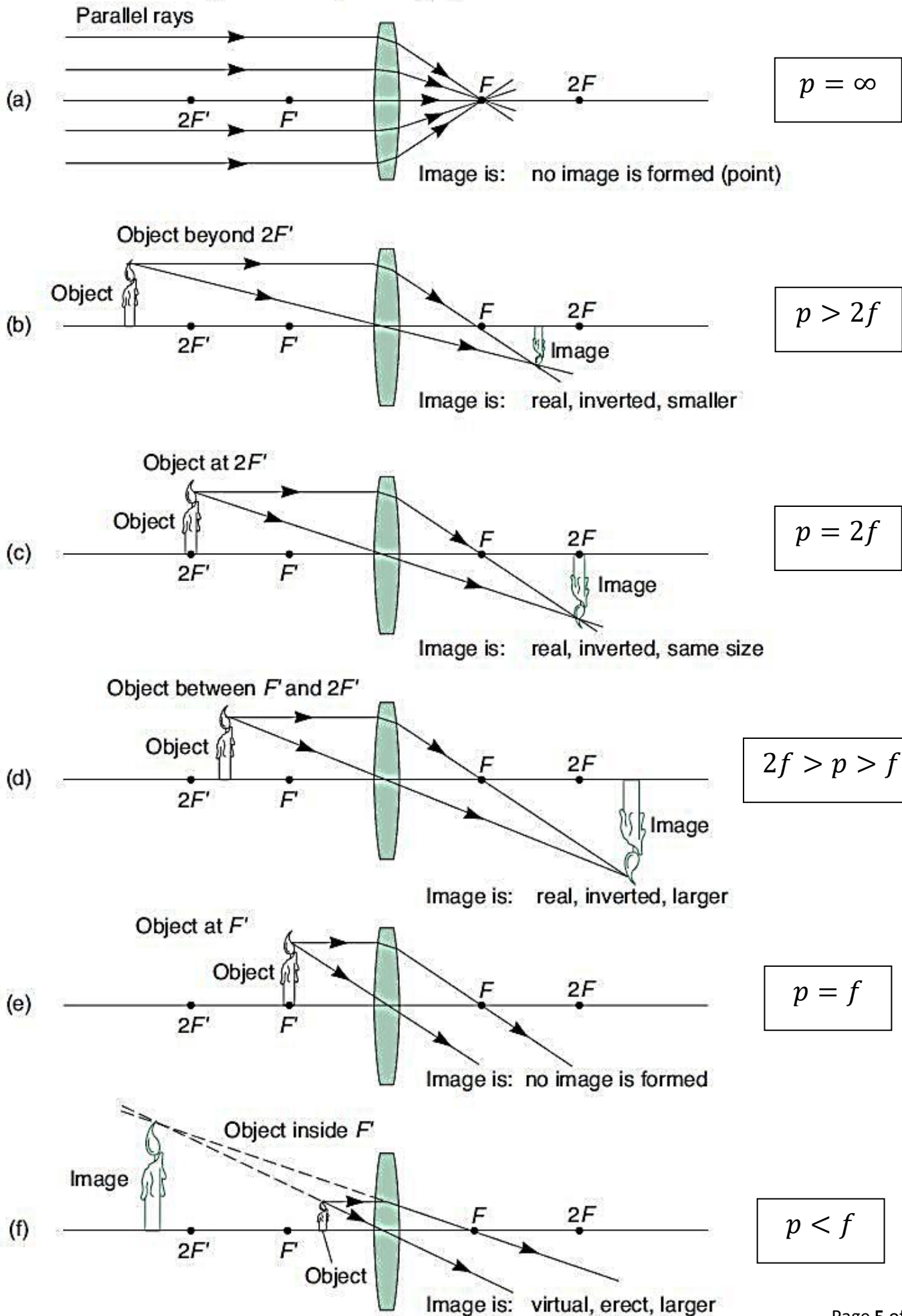


If $p > f$ real, inverted

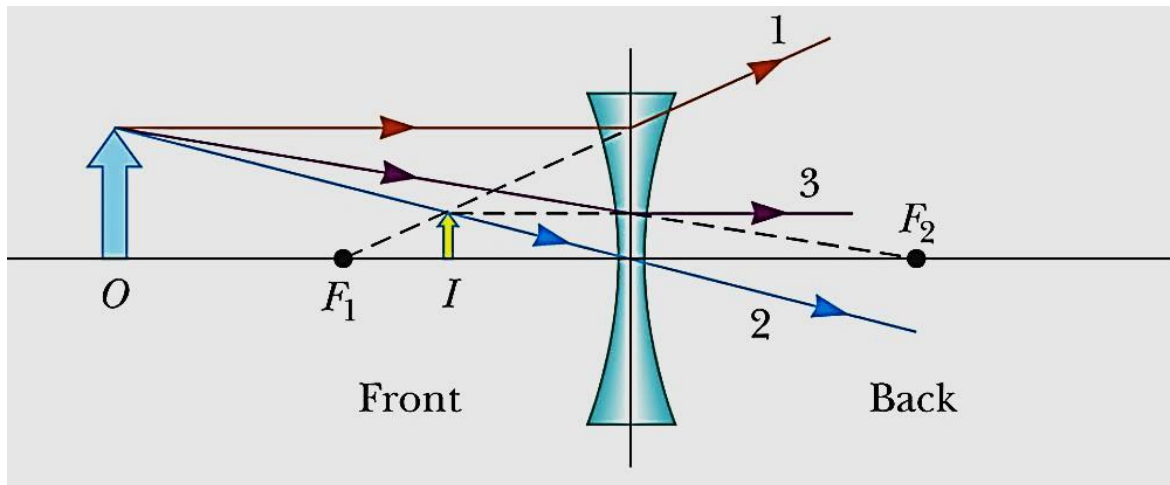


If $p < f$ virtual, upright

Images formed by converging lenses



- For Concave lens:



virtual, upright, reduced

- Lens maker equation:

$$\boxed{\frac{1}{p} + \frac{1}{q} = \frac{1}{f}} \quad \text{Lens maker's equation}$$

Magnification:

$$\boxed{M = -\frac{q}{p}}$$

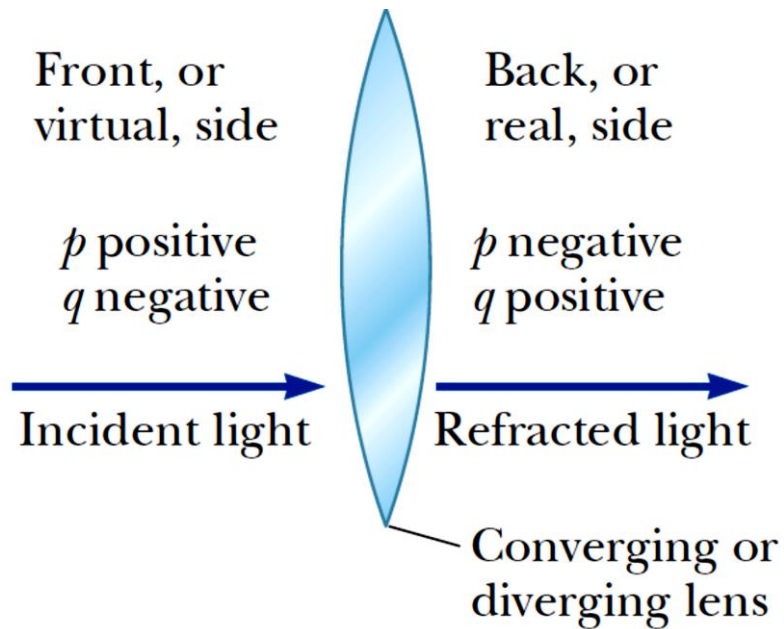
- Sign convection for thin lenses

Quantity	Positive when . . .	Negative when . . .
Object location (p)	object is in front of lens (real object).	object is in back of lens (virtual object).
Image location (q)	image is in back of lens (real image).	image is in front of lens (virtual image).
Image height (h')	image is upright.	image is inverted.
R_1 and R_2	center of curvature is in back of lens.	center of curvature is in front of lens.
Focal length (f)	a converging lens.	a diverging lens.

$$|M| = 1 \quad \text{image size} = \text{object size}$$

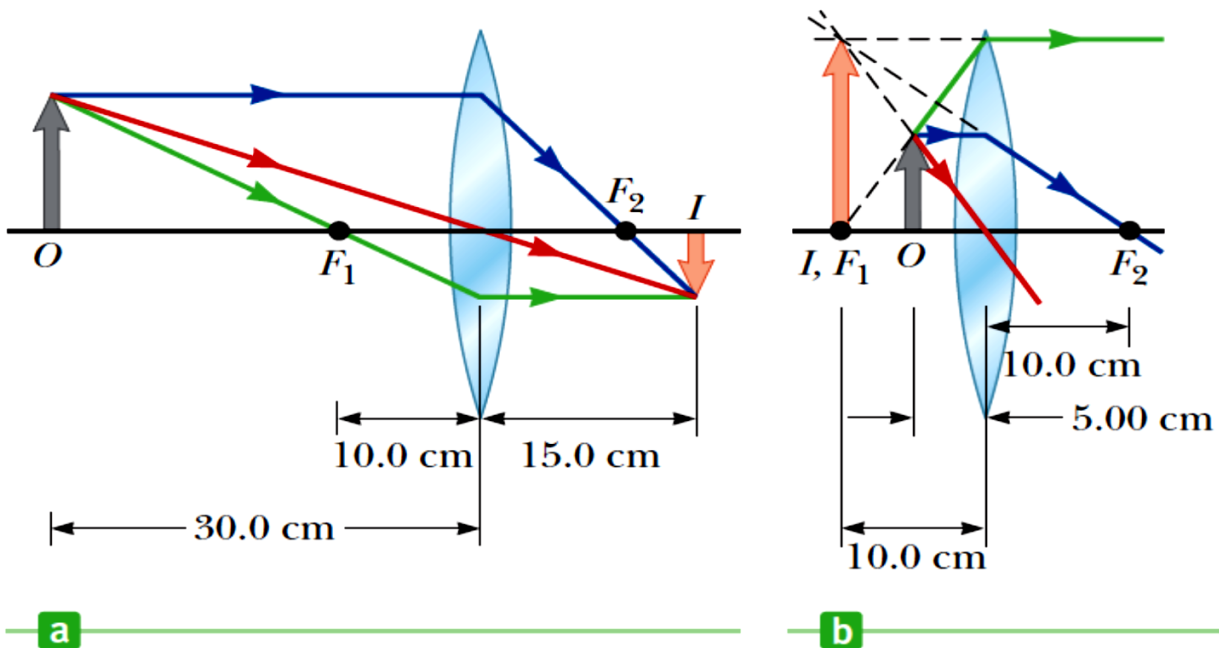
$$|M| > 1 \quad \text{image is enlarged}$$

$$|M| < 1 \quad \text{image is diminished}$$



- **Example 2:** A converging lens of focal length 10.0 cm forms images of an object situated at various distances.
 - If the object is placed 30.0 cm from the lens, locate the image, state whether it's real or virtual, and find its magnification.
 - Repeat the problem when the object is at 10.0 cm and (c) again when the object is 5.00 cm from the lens.

Solution



a)

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \longrightarrow \frac{1}{30} + \frac{1}{q} = \frac{1}{10}$$

$$q = +15 \text{ cm} \text{ (real image)}$$

$$M = \frac{-q}{p} = \frac{-15}{30} = -0.5 \text{ inverted \& reduced}$$

b)

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \longrightarrow \frac{1}{10} + \frac{1}{q} = \frac{1}{10}$$

$$q = \infty$$

c)

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \longrightarrow \frac{1}{5} + \frac{1}{q} = \frac{1}{10}$$

$$q = -10 \text{ cm} \text{ (virtual image)}$$

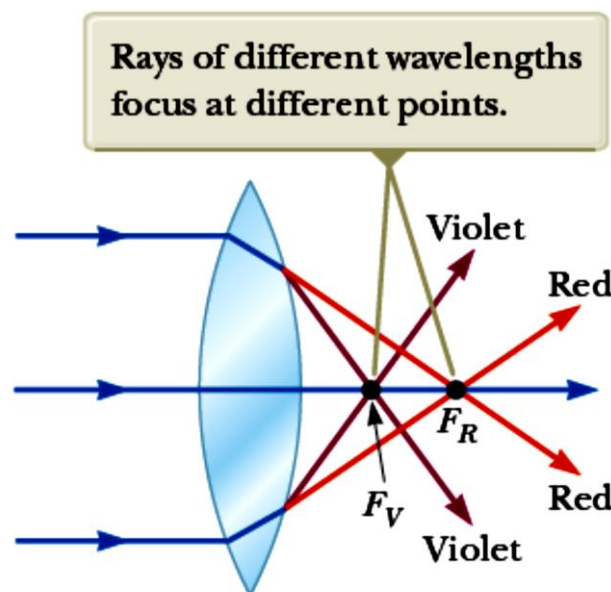
$$M = \frac{-q}{p} = \frac{-(-10)}{5} = 2 \text{ upright \& enlarged}$$

➤ Defects of the lenses:

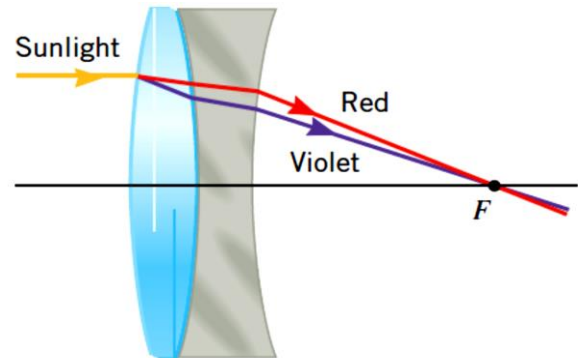
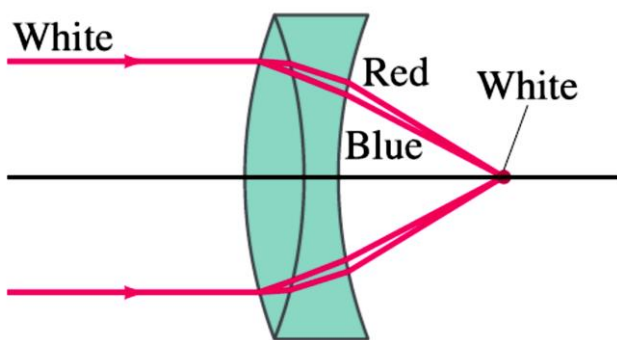
The aberrations of the lens causes decrease in the quality of the image

1) Chromatic aberration:

- It is the inability of the lens to converge or diverge all colours of the white light in one focus.
- A material's index of refraction varies with wavelength. Because of this phenomenon, violet rays are refracted more than red rays when white light passes through a lens. (The edges of the lens act like prism causing dispersion of white light)
- The focal length of a lens is greater for red light than for violet light.
- Other wavelengths have focal points intermediate between those of red and violet, which causes a blurred image.

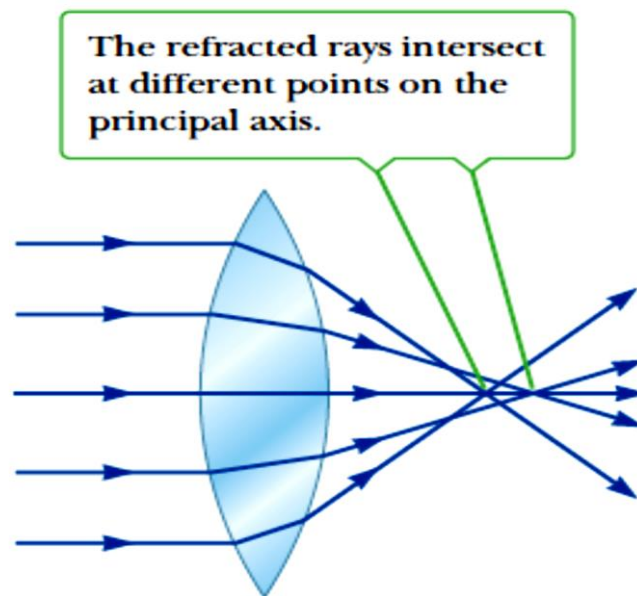


- To eliminate this defect we use Achromatic combination of lenses which is two lenses in contact but from different materials one of them is converging “convex” and the other is diverging “concave”.



2) Spherical aberration:

- It is the inability of the lens to converge or diverge all rays in one focus



- Spherical aberration results from the fact that the focal points of light rays passing far from the principal axis of a spherical lens (or mirror) are different from the focal points of rays with the same wavelength passing near the axis.
- To eliminate this defect, we can use a variable aperture. (barrier)

